

Vinh Ly

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Summary

Software Engineer with **18 years** of experience designing and shipping distributed backend systems at hypergrowth companies. Architected infrastructure at Google supporting **YouTube** and **Search** serving billions of users. Built real-time systems at **Whatnot** powering live commerce for millions of concurrent buyers. Delivered measurable outcomes including 40% reduction in P1 incidents and \$2M annual infrastructure savings. Led teams of 6 engineers and mentored 10+ to senior roles. Seeking to apply deep systems knowledge and security-minded approach to large-scale consumer infrastructure. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Systems & Infrastructure: distributed systems, backend services, Linux, networking, on-call /incident response, SRE

Cloud & DevOps: AWS, GCP, Kubernetes, CI/CD pipelines, monitoring, Terraform

Languages: Go, C/C++, Python, Java, SQL, TypeScript, JavaScript, HTML, CSS

Data & Storage: PostgreSQL, Redis, Bigtable, Spanner, Kafka

Work Experience

Staff Software Engineer | YouTube | 02/2025 - Present

- Collaborated with networking teams to optimize CDN routing algorithms, decreasing origin traffic by 25% and improving playback start times globally.
- Owned content delivery infrastructure serving 2 billion users, designing systems to handle 10M+ QPS with sub-100ms latency targets.
- Led migration of legacy video processing pipelines to Kubernetes-based microservices, improving throughput by 3x and reducing operational overhead.
- Established SLO frameworks and error budget policies that reduced customer-facing incidents by 30% quarter-over-quarter.
- Mentored 4 engineers on distributed systems design patterns and conducted architecture reviews for cross-functional teams.

Staff Software Engineer | Whatnot | 05/2024 - 02/2025

- Led migration of monolith to 15 microservices, reducing deployment time from 45 minutes to under 5 minutes using **CI/CD pipelines**.
- Built real-time bidding and inventory systems for live commerce using **Go** and **PostgreSQL**, supporting over 100k concurrent connections.
- Partnered with product and data teams to optimize listing quality algorithms, improving GMV by 15% through reduced drop-off rates.
- Owned on-call for platform reliability, achieving 99.99% uptime through automated failover and capacity planning across **Kubernetes** clusters.
- Established reliability standards and SLO frameworks that decreased P1 incidents by 40% across the platform.

Staff Software Engineer/Manager | Google | 03/2015 - 04/2024

- Established on-call rotation and incident management processes that decreased mean time to resolution from 45 minutes to under 12 minutes.
- Managed team of 6 engineers owning **YouTube** backend infrastructure serving over 2 billion users with 99.99% availability requirements.
- Mentored 10+ engineers with 3 promoted to Senior level and 2 to Staff level under direct guidance.

- Designed and built real-time recommendation and notification systems handling 10M+ QPS using **C++** and **Bigtable**.
- Reduced infrastructure costs by \$2M annually through capacity optimization, caching strategy improvements, and eviction policy tuning.

Software Engineer | Google | 02/2011 - 02/2015

- Developed **backend services** for **YouTube** search ranking infrastructure using **C++** and **Python**, improving relevance metrics by 12%.
- Optimized **distributed systems** query execution paths, reducing p99 latency from 800ms to 200ms for video metadata lookups.
- Built data pipelines processing billions of daily events using **MapReduce** and **BigQuery** for analytics and experimentation.
- Participated in on-call rotation for critical infrastructure serving global traffic with strict availability SLAs.

Software Design Engineer II | Microsoft | 10/2007 - 12/2011

- Developed **Windows** kernel components and system services using **C++**, shipping features in Windows 7 and Windows Server 2008 R2.
- Analyzed driver loading performance bottlenecks through kernel-level profiling, implementing optimizations that reduced boot time by 20%.
- Collaborated cross-functionally with hardware partners and product teams to ensure driver compatibility and reliability standards.
- Debugged complex kernel-mode crashes and memory corruption issues using **WinDbg** and custom instrumentation tools.

Software Design Engineer Intern | Microsoft | 06/2006 - 08/2006

- Developed automated testing framework for **Windows** UI components using **C++** and **MFC**, adopted by 5 product teams.
- Created test automation scripts that reduced manual testing cycle time by 60% for the Windows Desktop team.

Education

B.S. Honors Degrees in Computer Science

Virginia Tech | 2004 - 2007